



Introduction to p3d

Source: [vignettes/p3d-intro.Rmd](#)

Introduction

The p-value is a continuous measure of statistical evidence, yet in practice it is often reduced to a binary decision: $p < 0.05$ is “significant”, $p \geq 0.05$ is “not significant”. This dichotomous thinking is a major contributor to the reproducibility crisis in science.

The **p3d** package visualizes the p-value of a single-sample t-test as a **three-dimensional surface** over two parameters:

- **Effect size** (d , Cohen’s d)
- **Sample size** (n)

This vignette demonstrates the core functionality of the package.

The p-value surface

The central function of the package is `p_surface()`. Given a range of effect sizes and sample sizes, it computes the p-value for each combination:

```
surface <- p_surface(  
  d = seq(-2, 2, 0.1),  
  n = seq(5, 200, 5)  
)
```

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```
#> $ d      : num [1:41] -2 -1.9 -1.8 -1.7 -1.6 -1.5 -1.4 -1.3 -1.2 -1.1 ...
#> $ n      : num [1:40] 5 10 15 20 25 30 35 40 45 50 ...
#> $ p      : num [1:41, 1:40] 0.0111 0.0132 0.0158 0.0191 0.0232 ...
#> $ d_crit: num [1:40] 1.242 0.715 0.554 0.468 0.413 ...
#> $ alpha : num 0.05
#> $ grid  : 'data.frame': 1640 obs. of  5 variables:
#>   ..- attr(*, "out.attrs")=List of 2
```

The result is a list with several components:

- `d` : the vector of effect sizes
- `n` : the vector of sample sizes
- `p` : a matrix of p-values (dimensions $\text{length}(d) \times \text{length}(n)$)
- `d_crit` : critical effect sizes at $\alpha = 0.05$ for each n
- `alpha` : the significance level
- `grid` : a tidy data frame with all combinations

Inspecting the results

For example, the p-value at $d = 0.5, n = 30$ is:

```
surface$p[surface$d == 0.5, surface$n == 30]
#> [1] 0.01043739
```

The critical effect size at $n = 30$ is:

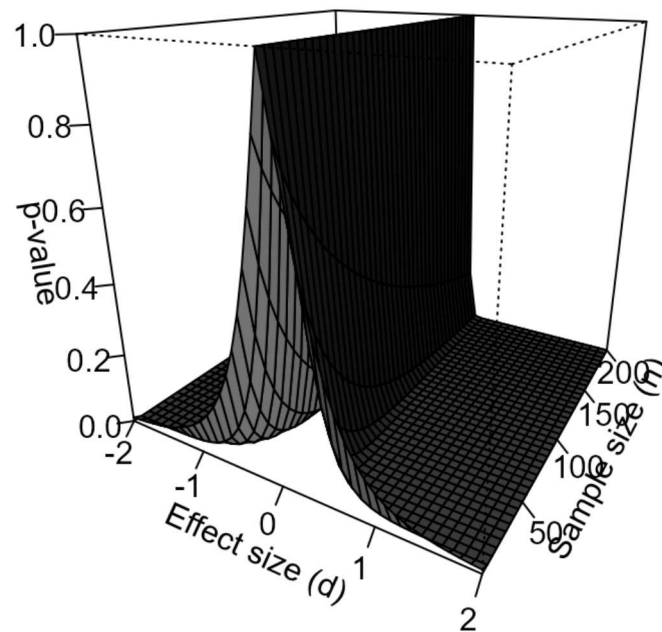
```
surface$d_crit[surface$n == 30]
#> [1] 0.3734061
```



Static 3D visualization

The simplest way to visualize the surface is `plot_p_static()`, which uses base R graphics and has **no external dependencies**:

```
plot_p_static(surface)
```



This is ideal for PDF output, journal figures, and printed reports.

Interactive 3D visualization

For interactive exploration, use `plot_p_surface()` with the `plotly` engine:

```
plot_p_surface(surface, engine = "plotly")
```

Alternatively, the `rgl` engine provides a native R window:



Both engines allow rotation, zooming, and panning.

2D comparison

For comparison with existing 2D packages such as `pvaluefunctions`, use `p_curve_2d()`:

```
curve_30 <- p_curve_2d(d = seq(-2, 2, 0.1), n = 30)
head(curve_30)
#>      d          p significant
#> 1 -2.0 8.021139e-12      TRUE
#> 2 -1.9 2.650857e-11      TRUE
#> 3 -1.8 9.099432e-11      TRUE
#> 4 -1.7 3.247269e-10      TRUE
#> 5 -1.6 1.205427e-09      TRUE
#> 6 -1.5 4.654771e-09      TRUE
```

The 2D curve is mathematically equivalent to a slice of the 3D surface at fixed n .

Real-data example

Let's apply `p3d` to a real dataset. The `mtcars` dataset contains fuel consumption (mpg) for 32 automobiles. Suppose we want to test whether the mean mpg is different from 20:

```
x <- mtcars$mpg
d_obs <- (mean(x) - 20) / sd(x)
n_obs <- length(x)

cat("Observed effect size:", round(d_obs, 4), "\n")
#> Observed effect size: 0.015
```

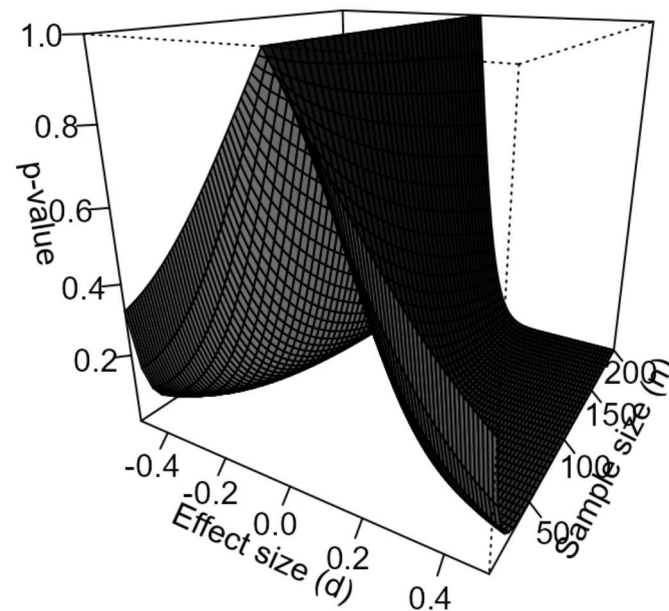


The p-value for this single observation is:

```
2 * (1 - pt(abs(d_obs) * sqrt(n_obs), df = n_obs - 1))  
#> [1] 0.9327606
```

By placing this observation on the 3D surface, we can see how the same effect size would behave at different sample sizes:

```
surface_real <- p_surface(  
  d = seq(-0.5, 0.5, 0.01),  
  n = seq(5, 200, 5)  
)  
  
plot_p_static(surface_real, theta = 30, phi = 20)
```



Conclusion

The **p3d** package provides a three-dimensional perspective on the p-value. By visualizing the p-value simultaneously across effect size and sample size, it reveals:

1. The **continuous** nature of the p-value (it is not a binary switch)
2. The **moderating effect** of sample size on statistical significance
3. The **interaction** between effect size and sample size



References

- Helske, J., Helske, S., & Cooper, M. (2021). *Visualizing Hypothesis Tests in Practice*. The R Journal.
- Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates.